

AI Software Development Bootcamp

36 WEEKS | ONLINE | \$14,995

Why AI Software Development?

Software development is entering a new era. The developers leading are the ones using AI to build more, ship faster, and solve larger problems with the same core engineering fundamentals. AI has become a standard layer in the development stack—not only as a feature inside products, but as a new class of engineering tools that augment how developers design, prototype, test, and deploy software. Developers who understand how to build with AI can expand the scope and speed of what they build and position themselves at the frontier of this transformation.

The most valuable developers will combine strong software engineering fundamentals with the ability to translate real-world problems into reliable, production-ready systems using AI-enabled workflows while maintaining quality, reliability, and sound engineering principles. Our AI Software Development Bootcamp is designed for this new reality. You will build strong foundations in software engineering while developing AI-focused skills across three areas:

- AI-native development environments (where AI is a primary interface to the codebase)
- AI application development and LLM engineering (integrating models through APIs, retrieval workflows, and designing AI-native product experiences)
- A foundation in machine learning engineering (core concepts, evaluation fundamentals, and the fundamentals of deploying models into production systems)

From building modern front-end interfaces to engineering backend APIs—including algorithms and data structures, AI-powered application development, and cloud deployment—you will develop the technical breadth and adaptability of modern software professionals.

Graduates leave the program prepared not only to write high-quality code, but also to design, build, and deploy systems that incorporate AI capabilities—equipping them with the practical skills to work in modern engineering environments using AI-enabled development tools and building AI-enabled products, whether in startups, technology companies, freelance work, or their own ventures.

Common Job Titles

- | | |
|----------------------------|------------------------|
| ✓ Software Engineer | ✓ Back End Developer |
| ✓ Software Developer | ✓ Full Stack Developer |
| ✓ Front End Developer | ✓ Mobile Engineer |
| ✓ DevOps Engineer | ✓ Systems Engineer |
| ✓ AI Application Developer | ✓ Technical Founder |



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Program Overview

Circuit Stream develops skill-based education around a simple principle: when the industry changes, training should change with it. The AI Software Development Bootcamp is a 36-week, career-focused program designed to prepare you for modern software engineering with an AI-enabled skill-set. Over the course of the program, you will develop strong software engineering foundations while learning to build full-stack applications across front-end, back-end, APIs, cloud deployment, and modern development workflows. You will also learn to work with AI-native development environments and practical AI application patterns that are part of how the best software teams design, build, and ship products.

This reflects a broader shift in the industry towards developers who can translate ideas into reliable, useful systems, combining strong engineering fundamentals with modern tooling and the ability to execute. That is the direction this program is built around: not theory, but practical capability and real implementation.

Throughout the bootcamp, you will explore how software engineers collaborate across product, design, and technical teams to bring complex systems to life. You will learn how modern development teams operate, how software moves from concept to production, and how AI-enabled workflows can expand what a skilled developer is able to build. You will actively use the latest tools like Cursor, Codex and Claude Code.

By the end of the program, you will complete a capstone project that demonstrates your ability to architect and develop a software solution using modern engineering principles. The goal is not only to understand how software is built, but to leave the program having deployed multiple real projects yourself - demonstrating your technical abilities and ability to adapt in the direction technology and software are heading.

What to Expect

Overall Expectation

- ✓ 36 weeks
- ✓ 280 hours of live sessions
- ✓ Up to 10 one-on-one career sessions
- ✓ Up to 6 projects for your portfolio
- ✓ Capstone project

Weekly Program

- ✓ 5-hours of live class time per week
- ✓ 2.5-hours of Technical and Career Labs every other week
- ✓ 10 hours of independent study

Experience

Certification

Upon successfully completing the AI Software Development Bootcamp, you will receive a Professional Software Development micro-credential. This attests to your proficiency and ability to engineer comprehensive and scalable software systems using industry-standard tools and methodologies.



Is This Course for You?

This foundational course is an excellent starting point for those aiming to establish a career in software development, offering an in-depth introduction to the industry's core practices and emerging trends.

AI Software Development Bootcamp is tailored for beginners, however, fundamental computer literacy skills are essential for a smooth learning experience. This encompasses file organization and the ability to navigate your operating system, be it Windows, macOS, or Linux.

Admission Process



STEP 1

Submit Online Application

Fill out our online application form and we'll get back to you as soon as possible.

[Submit here](#)



STEP 2

Program Acceptance

Your application will be reviewed by the Circuit Stream admissions team. Once reviewed, if you are accepted, you will receive an email officially offering you a seat in the program.

Outcomes

Software Engineering Foundations

- ✓ Start with web development fundamentals using HTML and CSS, and progress to JavaScript for dynamic interaction, preparing you for the complexities of software engineering, including best practices for code organization and responsive design.
-

Front-End Development & Full-Stack Integration

- ✓ Dive into modern front-end frameworks like React to create interactive user interfaces. Integrate these with backend APIs, learning full-stack development through practical projects that simulate real-world applications.
-

Backend Proficiency and DevOps

- ✓ Delve into server-side programming with Node.js, manage databases effectively, and understand back-end security. Incorporate DevOps practices, including server management and deployment with Docker, to streamline the software delivery process.
-

Artificial Intelligence and Neural Architectures

- ✓ Build a strong understanding of modern AI systems, from neural networks and deep learning fundamentals to transformers and large language models. Dive into navigating gradient-based optimization, mastering the art of prompt engineering, and architecting autonomous AI agents capable of complex reasoning and tool integration.
-

Mobile Development and Cross-Platform Innovation

- ✓ Expand your skills into mobile app development with React Native, adapting web principles for cross-platform applications. Learn platform-specific design, deployment, and accessibility best practices to build inclusive, user-friendly mobile experiences.
-

Algorithms and Structural Foundations

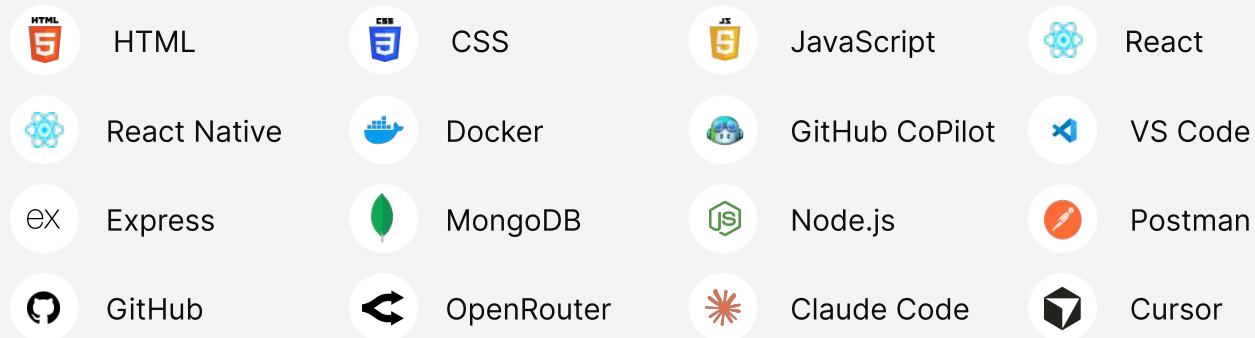
- ✓ Move beyond basic syntax to command the structural foundations that power world-class software. You will navigate the transition from linear arrays and sorting logic to the complex world of non-linear architectures like trees and graphs.
-

Career Development in Software Engineering

- ✓ Finding a job in a new industry is challenging and it requires the development of many soft skills. This course will provide guidance into the abilities and actions you need to confidently compete for a job.

Experience with Industry-Recognized Tools

AI Software Development Bootcamp will expose you to popular developer tools as you build out your software apps.



Program Structure

Overview



Live Sessions

Each week, there are five hours of live online teaching per week, split into two 2.5 hour sessions. Led by an instructor, you will apply a project-based methodology that gives you engaging challenges and hands-on problem-solving experience.



Lab Sessions

Labs are workshops where you practice and refine your skills through themed technical projects and case studies, career workshops, and dedicated working sessions.



Premium Support

Stay on track with access to premium academic support. A qualified expert will be available to address any technical questions you have and provide guidance as you work through your course material.



Career Services

Career coaching is at the heart of the program. From day one, we'll be there to provide guidance through career coaching, group workshops, one-on-one sessions, and guest lectures. The career component of the program runs concurrently with the technical training.

Learning Activities and Projects

AI Software Development Bootcamp is structured around project-based learning that has you gain practical skills while working on real-world projects to help build solid foundations for a career in software development. The course consists of:

01 Assignments

02 In-class challenges

03 Portfolio projects

04 Capstone project

Completion Requirements

To successfully complete the Bootcamp and earn your certification, you must finish with a grade of at least 70%.

● Course Assignments	-----	20%
● Midterm Project	-----	30%
● Capstone project	-----	50%

Prerequisites

AI Software Development Bootcamp is beginner-friendly and requires no prior experience to be admitted. Students can enter the program and be confidently learning from day one. To be accepted into the program, you must:

- Be 18 years of age or older.
- Have basic computer literacy.
- Have internet access.

Hardware Requirements

- Operating System: Windows 10 (64-bit) or later, or macOS Big Sur or later
- Memory: 8GB RAM minimum (16GB recommended)
- Storage: 256GB SSD minimum
- Processor: Intel Core i5 or higher, or Apple M1/M2 chip

Curriculum

Week 1 to 3

Foundations of Software Development

Begin your journey into software development by mastering the essentials of web development. You'll learn to construct well-structured web pages using HTML and to style them beautifully with CSS. This foundational knowledge sets the stage for you to create engaging and responsive user interfaces, understanding how each line of code translates into visual elements on the screen.

Key Outcomes

- ✓ Gain proficiency in HTML and CSS to develop structured and visually appealing websites
- ✓ Understand responsive design principles to ensure your web pages look great on any device
- ✓ Employ best practices for web development, from code organization to optimizing for performance and accessibility

Week 4 to 7

JavaScript

Starting with fundamental programming constructs, you'll quickly progress to creating rich, interactive user experiences. You'll employ JavaScript to respond to user interactions, manipulate the DOM in real-time, and make web pages react instantly to each user's input.

Key Outcomes

- ✓ Implement JavaScript to create interactive and dynamic web experiences
- ✓ Manipulate the DOM and harness the power of asynchronous JavaScript for responsive user interface
- ✓ Utilize AJAX, JSON, and third-party APIs to enrich your applications with dynamic content and seamless data integration

Back-End Development

Venture into the server-side realm with back-end development, where you'll build the engine room of your applications. Learn to set up servers, manage databases, and create APIs using Node.js. This module emphasizes the importance of security, teaching you to safeguard your applications against potential threats and vulnerabilities.

Key Outcomes

- ✓ Acquire the skills to build and maintain the back-end infrastructure using Node.js
- ✓ Implement security best practices to protect user data and ensure the integrity of your applications
- ✓ Develop and manage APIs that serve as vital conduits for front-end and back-end communication
- ✓ Leverage AI-powered tools to support common back-end development tasks and improve development workflows

Foundations of Artificial Intelligence

Dive into the core concepts powering today's most advanced AI systems. You'll explore neural networks and deep learning fundamentals, understand how models train through gradient descent and backpropagation, and examine transformer architectures behind large language models.

Key Outcomes

- ✓ Understand the mathematical and architectural foundations of neural networks, deep learning, and transformers
- ✓ Apply gradient-based optimization concepts, tokens, and embeddings in real-world AI workflows
- ✓ Design, prompt, and extend large language models using APIs, RAG, fine-tuning, structured outputs, and agent-based systems

Full Stack Integration

The full-stack integration module is where the front-end and back-end come together. You'll learn to weave front-end displays with back-end logic seamlessly, creating a cohesive and functional application. This module also covers the implementation of user authentication and session management, essential for personalized user experiences.

Key Outcomes

- ✓ Integrate front-end and back-end technologies to develop complete applications
- ✓ Manage user authentication and session states to provide secure and customized experiences
- ✓ Apply industry-standard tools and Agile methodologies to collaborate effectively in team environments
- ✓ Use AI-powered IDE tools to support API design, testing, and security best practices

Algorithm and Structural Foundations

Strengthen your software engineering skills with an in-depth look at algorithms and data structures. You'll learn how to organize data for optimal efficiency and solve complex problems systematically. This knowledge is crucial for writing efficient code that forms the backbone of powerful software solutions.

Key Outcomes

- ✓ Master the use of algorithms and data structures to solve complex problems with elegance and efficiency
- ✓ Utilize design patterns and SOLID principles to write clean, maintainable code
- ✓ Prepare for technical interviews with a solid understanding of computer science fundamentals

Front-End Development

This module introduces you to the dynamic world of front-end frameworks, focusing on React. You'll learn how to build dynamic, single-page applications (SPAs) and create reusable components that bring modern web applications to life. By understanding state management and lifecycle methods, you can construct interactive and data-driven user experiences.

Key Outcomes

- ✓ Create sophisticated user interfaces using React
- ✓ Understand the principles of component-based architecture to write reusable and maintainable code
- ✓ Employ testing strategies to ensure the reliability and robustness of your applications
- ✓ Leverage AI-powered IDE tools for front-end routing, debugging, and performance optimization

Deployment

Learn the art of deploying software and managing live applications. This module covers server management, continuous integration/continuous deployment (CI/CD) practices, and containerization with Docker. You'll understand how to get your applications out of development and into the hands of users reliably and efficiently.

Key Outcomes

- ✓ Deploy web applications using modern DevOps tools and techniques
- ✓ Configure servers and set up automated deployment pipelines for streamlined software delivery
- ✓ Use Docker to create containerized environments, ensuring consistency across development and production
- ✓ Leverage AI-powered tools and autonomous agents (e.g., GitHub Copilot, Claude Code) to streamline production workflows

Week 30

AI Systems in Production

The module focuses on deploying intelligent applications in real-world environments. You'll build an end-to-end Retrieval-Augmented Generation (RAG) system and learn how to move AI solutions from prototype to production.

Key Outcomes

- ✓ Design and implement an end-to-end RAG pipeline integrating data retrieval and large language models
- ✓ Apply tracing, monitoring, and evaluation frameworks to measure system performance and reliability
- ✓ Implement guardrails and production best practices to deploy safe, scalable AI systems

Week 31 to 33

Mobile Development

Embrace the mobile-first approach as you learn to translate your web development skills to the mobile platform. Focusing on React Native, you'll adapt your applications for iOS and Android, ensuring they are responsive, intuitive, and tap into the unique features of mobile devices.

Key Outcomes

- ✓ Develop cross-platform mobile applications that offer a seamless user experience across devices
- ✓ Utilize mobile-specific features and design patterns to enhance the functionality and usability of your applications
- ✓ Navigate the ecosystem of mobile app development, from platform guidelines to app store distribution
- ✓ Implement accessibility features in mobile applications, ensuring inclusivity and usability for all users, regardless of their abilities.

Capstone Project

Over 4 weeks, you'll harness your comprehensive AI software engineering skills to identify a real-world problem and craft a targeted software solution. Your journey culminates in a presentation to industry experts, showcasing your ability to transform challenges into impactful technology solutions. This is where you transition from learning to leading, demonstrating your readiness for the professional tech landscape.

Project Breakdown

Foundations & JavaScript	Create a personal portfolio website that showcases your work, skills, and potential. In the JavaScript unit, add interactivity to the website.
Back-End Development	Develop a RESTful API for a task management application using Node.js. Implement CRUD operations and ensure data is securely stored and retrieved from a database.
Foundations of Artificial Intelligence	Apply core AI concepts by building neural network workflows, working directly with large language models through APIs, and designing prompts, RAG pipelines, and AI agents to solve structured problems.
Full-Stack Integration	Connect a provided user interface to a custom API to enable user account creation, profile editing, and data storage.
Algorithm & Structural Foundations	Develop a dynamic data dashboard. Use HTML, CSS, and JavaScript, alongside visualization libraries like D3.js, to present complex datasets through interactive charts and graphs.
Front-End Development	Develop a weather forecast application where users can enter a location and receive a weekly weather outlook. Ensure functionality with comprehensive test coverage.
Mobile Development	Convert an existing web application into a mobile app using React Native, maintaining feature parity and seamless data synchronization between desktop and mobile versions.

Career Development Program

During our career development program, you will be equipped to showcase your software development skills and behavioral strengths, craft a captivating portfolio featuring your AI Software Development Bootcamp projects, and elevate your digital presence to align with your personal brand.

Career Development Program Highlights

- ✓ Write resumes and cover letters
- ✓ Craft a standout personal brand
- ✓ Build an industry-ready portfolio
- ✓ Master interviews with hands-on practice
- ✓ Network with mentors and industry experts
- ✓ Present your project to industry partners

Units

Goal Setting

- ✓ Turn your ambitions into action. Learn to define clear career goals, map the skills you need, and navigate the job market to create a practical roadmap for success. By the end, you'll have a personalized career plan with milestones to guide your journey.

Digital Presence

- ✓ Build a personal brand that stands out online. Learn to craft your story, showcase your skills, and create a professional digital presence across your portfolio and social platforms — setting the stage for networking and career opportunities.

Portfolio

- ✓ Showcase your skills with impact. Learn to tell compelling stories around your projects using the STAR method and build a professional portfolio that highlights your abilities and personal brand to potential employers.

Resume and Cover Letter

- ✓ Craft applications that get noticed. Learn to tailor your resume and cover letter to specific jobs, highlight your skills effectively, and align your LinkedIn profile — all while building a strategic job search plan.

Interviews

- ✓ Ace your interviews with confidence. Practice technical, behavioral, and screening questions through realistic mock sessions, learning to communicate your skills and ideas clearly to industry professionals.

Pitch Day Prep

- ✓ Polish your pitch and showcase your work. Prepare your presentation, refine your elevator pitch, and get expert feedback to confidently present your Capstone project to industry partners, making connections that can jumpstart your career.

Networking Events



Pitch Day

You will have the opportunity to take part in Pitch Day, a day dedicated to showcasing your capstone project that takes place a month after Bootcamp ends. The event ranges between 2-3 hours, and you will have the opportunity to present and pitch your project to hiring managers and industry experts. This is also an opportunity to gain feedback from peers and participants alike.



Ask Me Anything

Our Virtual Career Ask Me Anything (AMA) events connect students and prospective learners with industry professionals and alumni in a live, interactive setting. Hear firsthand about real career paths, day-to-day work, and the skills needed to succeed in the industry. Ask questions in real time and gain practical advice on breaking into the field, preparing for interviews, and building your career.

Career Program Benefits

- ✓ 6 Career Labs (combination of live sessions and pre-recorded sessions)
- ✓ Technical Mock Interview with a Lead Instructor
- ✓ Real-Time Project Feedback
- ✓ Industry Mentors & Hiring Managers Introductions
- ✓ Career Support 180 Days After Graduation

Pricing

Pay Upfront

Pay in full and get a 15% discount. Circuit Stream accepts wire transfers and credit card payments.

\$14,995
excluding tax

Deferred Payments

Circuit Stream offers 3, 6, 12, and 24-month payment plans. Payment plans are subject to a one-time admin fee and a credit check during your application review.

for as low as
\$600/mo



International Students

For students outside of the U.S, Circuit Stream offers a 3-month payment plan.

for as low as
\$4,998/mo



Newcomers to Canada

Circuit Stream has partnered with Windmill to offer affordable career loans for skilled newcomers to Canada. More details here: <https://www.windmillmicrolending.org/our-loan>.



Ready to get started?

Fill out our online application form and we'll get back to you as soon as possible.

[Submit here](#)

Have questions?

Schedule a call with a Program Advisor to learn how this course can help you.

[Schedule here](#)

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